Triple Integral over a Paraboloid and a Plane

Stewart, *Calculus: Early Transcendentals*, Section 15.7, Exercise 17

Problem

Evaluate the triple integral.

$$\iiint_E x \, \mathrm{d}V,$$

where E is bounded by the paraboloid $x = 4y^2 + 4z^2$ and the plane $x = 4$.

Part (a)

This exercise has a single part. The whole solution is Part (a), and the supporting files are named with `part1`.

The symbols. E is the solid region of $\mathbb{R}^3$ that the two bounding surfaces enclose. $\mathrm{d}V$ is the volume element, that is, the volume of an infinitesimal piece of E. The symbol x, used both as the integrand and as one of the three coordinates, is the first coordinate of a point (x, y, z) of E. Because $\mathrm{d}V$ is positive and $x \geq 0$ on E, the integral is the volume of E weighted by x. The symbol D below denotes the projection of E onto the yz-plane. In that plane, r is the distance of a point from the origin, that is, $r^2 = y^2 + z^2$, and θ is the polar angle measured from the positive y-axis toward the positive z-axis. Then $\mathrm{d}A = r \, \mathrm{d}r \, \mathrm{d}\theta$ is the area element of the plane. This exercise needs no spherical coordinates (ρ, θ, φ); the symbol θ above is the polar angle in the yz-plane.

Step 1. Describe the solid

The paraboloid $x = 4y^2 + 4z^2$ opens toward the positive x-axis and touches the origin. The plane $x = 4$ cuts it off. The two surfaces meet where both equations hold:

$$4y^2 + 4z^2 = 4 \quad \Longleftrightarrow \quad y^2 + z^2 = 1. \tag{1}$$

Equation (1) is a circle of radius 1 in the plane $x = 4$. Therefore E lies over the closed disk

$$D = \big\{(y, z) : y^2 + z^2 \leq 1\big\} \tag{2}$$

of the yz-plane. Figure 1 shows the paraboloid, the cap on the plane $x = 4$, and the circle where the two surfaces meet. For every point (y, z) of D, the paraboloid gives the lower value $x = 4(y^2 + z^2)$ and the plane gives the upper value $x = 4$. Hence

$$E = \big\{(x, y, z) : y^2 + z^2 \leq 1, \ 4y^2 + 4z^2 \leq x \leq 4\big\}. \tag{3}$$

The difference of the two bounds is $4 - 4(y^2 + z^2)$, which is nonnegative on D and is zero only on the boundary circle. So E is a genuine solid, and it has the shape of a bowl that is filled up to the flat lid $x = 4$.

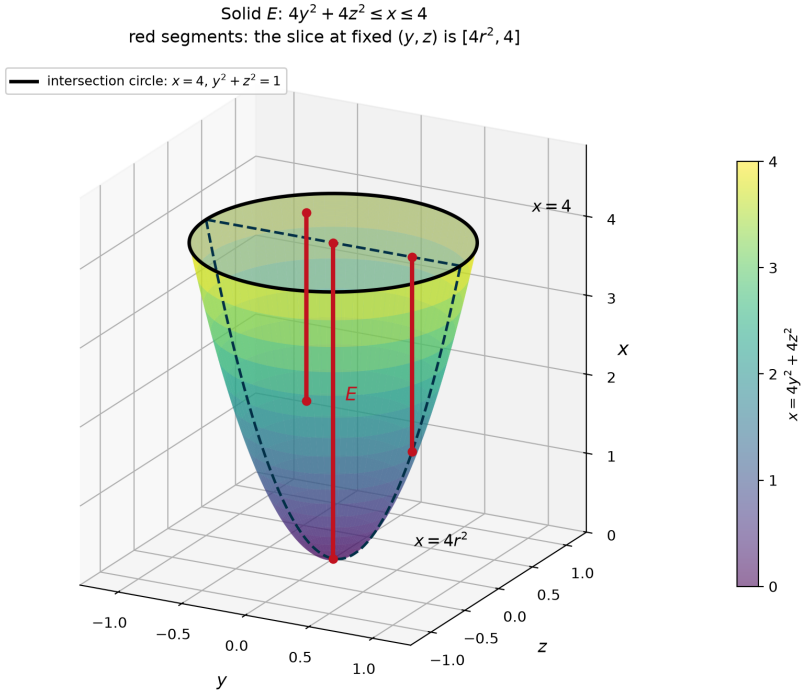


Figure 1: The solid E of Exercise 15.7.17. The curved surface is the paraboloid $x = 4y^2 + 4z^2$, drawn for $0 \le r \le 1$. The pale surface at the top is the part of the plane $x = 4$ that closes the solid. The heavy black curve is the intersection circle $x = 4$, $y^2 + z^2 = 1$. Each red segment is the slice of E at one fixed pair (y, z); the segment runs from $x = 4r^2$ up to $x = 4$. The dashed curve is the cross section of the paraboloid in the plane $z = 0$. The vertical axis of the picture is x.

Step 2. Choose the order of integration

Region (3) is simple in x. A line parallel to the x-axis through a point (y, z) of the interior of D enters E at $x = 4y^2 + 4z^2$ and leaves E at $x = 4$. It meets E in exactly one segment. Therefore we may integrate with respect to x first, and the outer double integral runs over the disk D:

$$\iiint_E x \, \mathrm{d}V = \iint_D \left(\int_{4y^2+4z^2}^{4} x \, \mathrm{d}x \right) \mathrm{d}A. \tag{4}$$

This order is legal by Fubini's theorem, because the integrand x is continuous on E and the bounds are continuous functions of (y, z) on D.

The remaining double integral runs over the disk D. The disk is described in polar coordinates by $0 \le r \le 1$ and $0 \le \theta \le 2\pi$, and the area element is $\mathrm{d}A = r \, \mathrm{d}r \, \mathrm{d}\theta$. So we also convert the outer integral to polar coordinates:

$$\iiint_E x \, \mathrm{d}V = \int_0^{2\pi} \int_0^{1} \left(\int_{4r^2}^{4} x \, \mathrm{d}x \right) r \, \mathrm{d}r \, \mathrm{d}\theta. \tag{5}$$

The order inside Equation (5) is x, then r, then θ. Figure 2 shows one slice at fixed y in the plane $z = 0$.

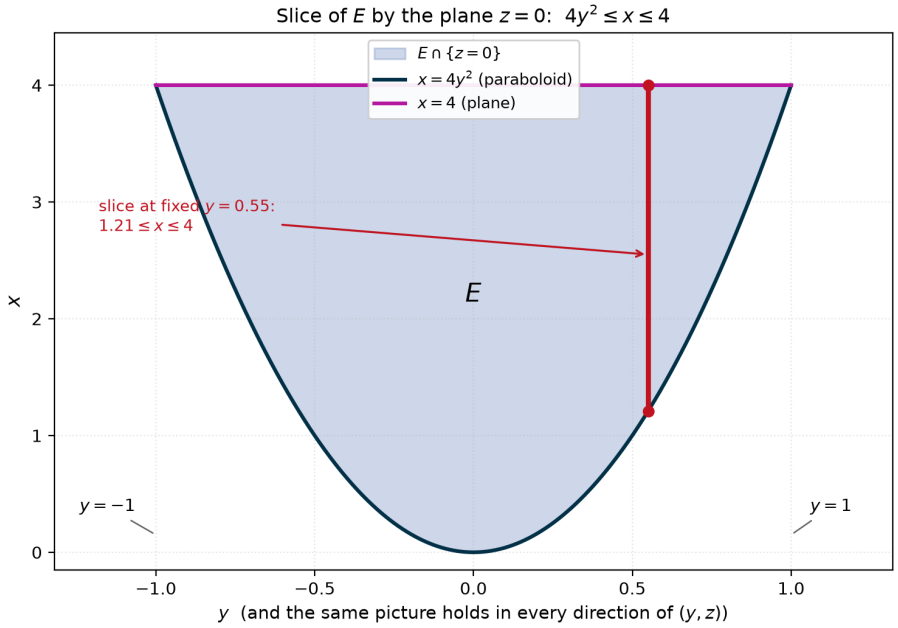


Figure 2: The cross section of E by the plane $z = 0$. The region lies between the parabola $x = 4y^2$ and the line $x = 4$, for $-1 \le y \le 1$. The red segment is the slice at $y = 0.55$; it runs from $x = 1.21$ to $x = 4$. A horizontal line at height x crosses the region in one interval, and a vertical line crosses it in one interval. Both facts show that the order x first is correct.

Step 3. Compute the inner integral

The antiderivative of x with respect to x is $x^2/2$. Hence

$$\begin{aligned}\int_{4r^2}^{4} x\,\mathrm{d}x &= \left[\frac{x^2}{2}\right]_{x=4r^2}^{x=4} = \frac{4^2}{2} - \frac{(4r^2)^2}{2} \\ &= \frac{16}{2} - \frac{16r^4}{2} = 8 - 8r^4. \end{aligned} \tag{6}$$

The result is nonnegative on $0 \le r \le 1$; it is equal to 8 at $r = 0$ and to 0 at $r = 1$. This agrees with the picture, because the vertical slice shrinks to a point at the rim of the bowl.

Step 4. Compute the polar integral

Insert Equation (6) into Equation (5). The integrand of the double integral becomes $(8 - 8r^4)r = 8r - 8r^5$.

$$\begin{aligned}\iiint_E x\,\mathrm{d}V &= \int_0^{2\pi}\int_0^1 (8r - 8r^5)\,\mathrm{d}r\,\mathrm{d}\theta = \int_0^{2\pi}\left[4r^2 - \frac{8r^6}{6}\right]_{r=0}^{r=1}\mathrm{d}\theta \\ &= \int_0^{2\pi}\left(4 - \frac{4}{3}\right)\mathrm{d}\theta = \int_0^{2\pi}\frac{8}{3}\,\mathrm{d}\theta = \frac{8}{3}\cdot 2\pi = \frac{16\pi}{3}. \end{aligned} \tag{7}$$

The antiderivative of $8r$ is $4r^2$, and the antiderivative of $8r^5$ is $8r^6/6$. The θ-integral of a constant over $[0, 2\pi]$ is that constant times 2π.

Answer

$$\boxed{\iiint_E x \, \mathrm{d}V = \frac{16\pi}{3} \approx 16.7552}$$

Step 5. Check the value

The value was checked twice. The first check is exact and uses SymPy. The following two computations give the same result:

- polar form, as above, with symbolic r and θ;
- a Cartesian order x, then y, then z with $y = \pm\sqrt{1 - z^2}$ and $-1 \le z \le 1$, which is a different description of the same solid.

The second check is numerical and uses NumPy only. It applies three independent methods: a two-dimensional composite Simpson rule on the polar integral, a Monte Carlo estimate over the box $[0, 4] \times [-1, 1] \times [-1, 1]$, and a midpoint voxel sum. Table 1 collects the numbers.

Table 1: Numerical confirmation of the exact value $16\pi/3$. The error column compares each estimate with 16.7551608191456.

Method	Estimate	Absolute error
Two-dimensional Simpson rule (400×400)	16.755160818491	6.5×10^{-10}
Monte Carlo, $3 \cdot 10^6$ samples	16.737315	1.8×10^{-2}
Midpoint voxel sum, 240^3 cells	16.751468171	3.7×10^{-3}
Exact value $16\pi/3$	16.755160819146	—

The three numerical estimates agree with $16\pi/3$ to the accuracy that each method can reach. The scripts `part1_step1.py` to `part1_step4.py`, `part1_figure1.py`, and `part1_figure2.py` produce every number and both pictures in this document, and each script ends with the line `PASS`.

Step 6. Alternative exact evaluation in Cartesian coordinates

The same value follows from the order x, then y, then z, with no polar coordinates at all. This second exact route is a useful check, because it uses a different antiderivative for the y-integration.

The inner integral is again Equation (6) with $r^2 = y^2 + z^2$, so the double integral over D is

$$\iint_D \bigl(8 - 8(y^2 + z^2)^2\bigr) \, \mathrm{d}A = \int_{-1}^{1} \int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} \bigl(8 - 8(y^2 + z^2)^2\bigr) \, \mathrm{d}y \, \mathrm{d}z. \tag{8}$$

The disk is symmetric in y, and the integrand is an even function of y. Expanding the square gives $8 - 8y^4 - 16y^2z^2 - 8z^4$, and term by term integration over the symmetric interval $[-\sqrt{1 - z^2}, \sqrt{1 - z^2}]$

gives

$$\int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} \left(8 - 8(y^2+z^2)^2\right) \mathrm{d}y = \frac{64}{15}\,(1-z^2)^{3/2}\,(2z^2+3). \tag{9}$$

Substitute $z = \sin\varphi$ with $\mathrm{d}z = \cos\varphi\,\mathrm{d}\varphi$. The interval $-1 \le z \le 1$ becomes $-\pi/2 \le \varphi \le \pi/2$, and $(1-z^2)^{3/2} = \cos^3\varphi$. The integrand is even in φ, so

$$\begin{aligned}\iiint_E x\,\mathrm{d}V &= \frac{64}{15}\int_{-1}^{1}(1-z^2)^{3/2}(2z^2+3)\,\mathrm{d}z = \frac{128}{15}\int_0^{\pi/2}\cos^4\varphi\left(2\sin^2\varphi+3\right)\mathrm{d}\varphi \\ &= \frac{128}{15}\int_0^{\pi/2}\left(5\cos^4\varphi - 2\cos^6\varphi\right)\mathrm{d}\varphi, \end{aligned}\tag{10}$$

where the last step uses $2\sin^2\varphi + 3 = 5 - 2\cos^2\varphi$. The reduction formula for powers of the cosine gives

$$\int_0^{\pi/2}\cos^4\varphi\,\mathrm{d}\varphi = \frac{3\pi}{16}, \qquad \int_0^{\pi/2}\cos^6\varphi\,\mathrm{d}\varphi = \frac{5\pi}{32}. \tag{11}$$

Therefore

$$\iiint_E x\,\mathrm{d}V = \frac{128}{15}\left(5\cdot\frac{3\pi}{16} - 2\cdot\frac{5\pi}{32}\right) = \frac{128}{15}\cdot\frac{5\pi}{8} = \frac{16\pi}{3}, \tag{12}$$

which is the same value as Equation (7).

How to solve problems of this kind

1. **Find the intersection of the surfaces.** Set the two bounding equations equal, as in Equation (1). The result is the curve where the solid closes, and it gives the rim of the projection.
2. **Project onto a coordinate plane.** Choose the plane that makes the projection simple. Here the paraboloid is written as $x = f(y,z)$, so the natural projection is the yz-plane, and the projection is the disk $y^2 + z^2 \le 1$.
3. **Test whether the solid is simple in the remaining variable.** Draw a line parallel to the remaining axis. If the line meets the solid in one segment for every interior point of the projection, then that variable is the inner variable. Here x is the inner variable, with bounds $4y^2 + 4z^2 \le x \le 4$.
4. **Match the coordinates to the projection.** A disk projection calls for polar coordinates in that plane. A circular cylinder calls for cylindrical coordinates. A spherical cap calls for spherical coordinates. Do not use spherical coordinates for a solid that is a box in some other system.
5. **Write the volume element of the chosen system.** For polar coordinates in the yz-plane the element is $r\,\mathrm{d}r\,\mathrm{d}\theta$; forgetting the extra factor r is the most common error in this type of problem.
6. **Integrate from the inside out.** Compute one antiderivative at a time. Check the sign at a few sample points, as in Equation (6), to catch a swapped pair of bounds.
7. **Check the answer.** Compare the exact result with an independent numerical method, and compare the magnitude with a rough geometric estimate. Here the volume of E is 2π, and the average value of x on E is close to 2.67, so an answer near 16.8 is reasonable.